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ME-1845

B. Tech. (First Semester)

EXAMINATION, 2025-26

ENGG. MECHANICS

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 8 marks. $7 \times 8 = 56$

~~X~~ Prove and explain parallelogram law of theorem.

P. T. O.

2. A smooth circular cylinder of radius 1.5 m is lying in a triangular groove, one side of which makes 15° angle and the other 40° with the horizontal. Find the reactions at the surfaces of contact, if there is no friction and the cylinder weighs 100 N.
3. What are various types of loadings ? Distinguish clearly between uniformly and non-uniformly distributed load.
4. What is truss ? Explain its governing equation.
5. What do you understand by centroid and centre of gravity ?
6. State and prove the parallel axis theorem of moment of inertia.
7. A stone is thrown vertically upwards with a velocity of 30 m/s from a top of a tower 3403 m high. Find the total time taken by the stone to reach the foot of the tower.

8. Explain of the following :

- (i) Velocity
- (ii) Acceleration
- (iii) Displacement

9. What do you understand by Lami's theorem ? Explain in detail.

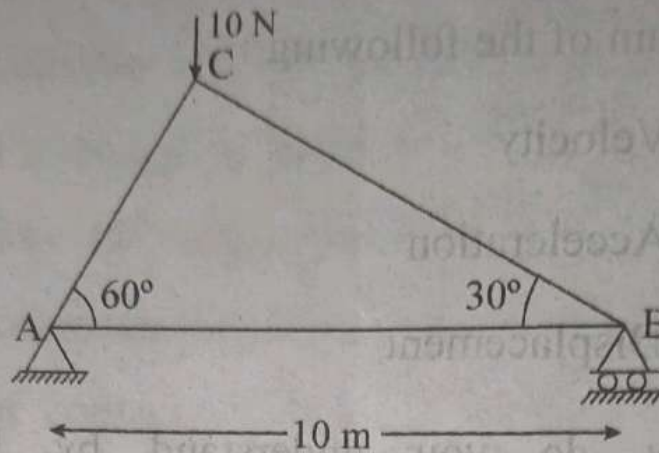
10. A hollow steel tube 3.5 m long has external diameter of 120 mm. In order to determine the internal diameter, the tube was subjected to a tensile load of 400 kN and extension was measured to be 2 mm. If the modulus of elasticity for the tube material is 200 GPa, determine the internal diameter of the tube.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

1. Find the force in members of truss.



2. Explain Newton's law of motion in detail with clear illustration.

X A lift goes up to a height of 900 m with a constant acceleration and then the next 300 m with is constant retardation and comes to rest.

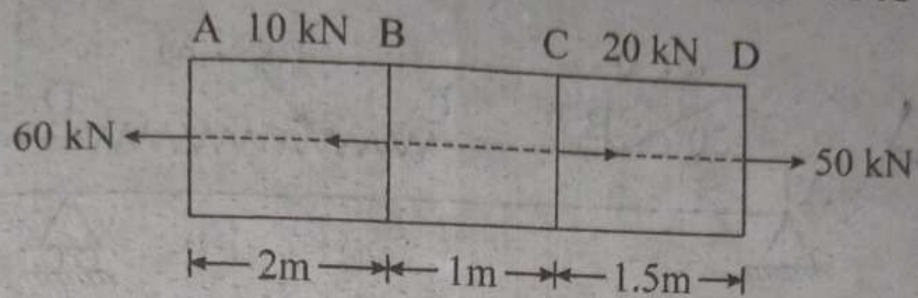
Find :

- (i) The maximum velocity of the lift if the time taken to travel is 30 seconds.
- (ii) Acceleration of the lift.

4. A steel rod ABCD 4.5 m long and 25 mm in diameter is subjected to the forces as shown in figure below. If the value of Young's modulus is 200 GPa determine the deformations.

[5]

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Section—C

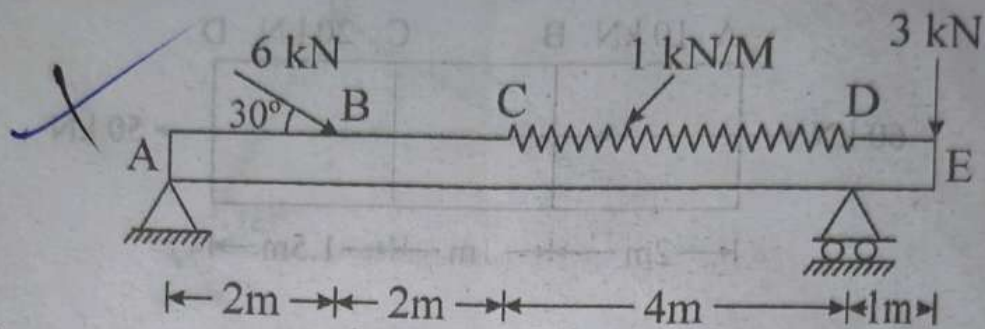
(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. An effort of 1200 N is required just to move a certain body up on inclined plane of 15° angle the force acting parallel to the plane. If the angle of inclination of the plane is made 20° the effort requires, again applied parallel to the plane, is found to be 1230 N. Find the weight of the body and the coefficient of friction.

✱ Analyse the beam shown in figure and draw the SFD and BMD. Locate the points of contraflexure if any.

P. T. O.



3. Find the moment of inertia of a T-Section with flange as 150 mm $\times$ 50 mm and web 150 mm $\times$ 50 mm about XX and YY axes through the centre of gravity of the section.
4. (a) Explain the friction in detail with examples.
- (b) Prove and explain the tension ratio between tight and slack side of belt and pulley.

$$\frac{T_1}{T_2} = e^{\mu\theta}$$

where all denotations have their usual meanings.

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